Table 1. AUROC ×100 (higher is better)

	Same-source			Different-source		
Method	**Alpaca**	**MMLU**	**Big-Math**	**Magpie**	**MMLU-Pro**	**MATH**
Strict 5,955 baseline — mean ± sample SD; repeat identity in notes						
SelfAsk	54.87 ± 0.00	54.14 ± 0.00	47.87 ± 0.00	44.08 ± 0.00	56.37 ± 0.00	44.14 ± 0.00
Entropy	49.78 ± 0.00	57.68 ± 0.00	48.61 ± 0.00	47.64 ± 0.00	57.11 ± 0.00	45.55 ± 0.00
MaxLogits	46.67 ± 0.00	56.07 ± 0.00	47.68 ± 0.00	40.54 ± 0.00	56.41 ± 0.00	43.04 ± 0.00
Conf. Margin	49.16 ± 0.00	53.75 ± 0.00	48.46 ± 0.00	47.00 ± 0.00	56.42 ± 0.00	41.97 ± 0.00
Emb. MLP	58.49 ± 0.66	63.69 ± 0.12	63.16 ± 0.60	72.19 ± 0.42	61.55 ± 0.30	65.82 ± 0.71
HybridLLM	63.70 ± 1.92	65.13 ± 1.29	64.21 ± 0.84	66.23 ± 8.05	60.56 ± 3.01	67.26 ± 2.39
RouteLLM	57.69 ± 0.43	58.47 ± 0.32	60.74 ± 0.06	74.40 ± 0.15	52.62 ± 0.44	61.74 ± 0.09
Jev	59.27 ± 0.08	69.77 ± 0.71	71.92 ± 0.57	70.59 ± 0.19	70.33 ± 0.22	80.47 ± 0.17
MeanProbe	65.71 ± 0.13	67.62 ± 0.11	67.76 ± 0.04	76.43 ± 0.11	64.66 ± 0.08	72.46 ± 0.18
StartLux — one exploratory run, no SD						
StartLux 0.8B Instant (Q)	58.34	57.17	62.51	65.05	58.71	72.37
StartLux 0.8B Instant (Q+A)	55.19	66.82	81.77	58.66	71.32	87.25
StartLux 2B Instant (Q)	57.43	49.26	59.74	64.22	53.76	72.53
StartLux 2B Instant (Q+A)	62.49	74.93	87.62	73.34	79.50	91.36
StartLux 4B Instant (Q)	66.59	67.30	70.70	76.02	67.44	80.91
StartLux 4B Instant (Q+A)	68.92	88.17	92.51	78.09	87.47	94.72
StartLux 9B Instant (Q)	68.06	67.80	72.94	76.56	65.19	79.33
StartLux 9B Instant (Q+A)	74.09	90.71	93.29	79.28	89.39	95.64
StartLux 27B Instant (Q)	66.83	74.90	76.43	73.91	71.72	82.92
StartLux 27B Instant (Q+A)	73.90	95.15	96.49	81.92	93.51	98.11
StartLux 9B AutoThink (Q)	65.31	64.48	72.56	76.24	61.51	79.18
StartLux 27B AutoThink (Q)	68.10	74.49	75.37	80.53	71.93	82.85

Strict clean 5,955 comparison: AUROC ×100 (higher is better). Baseline: 9 previous methods from the 5,955 clean sources; SD is across the actual repeats named in the table notes (training seeds, API executions or fixed executions). StartLux: one exploratory execution per condition (no SD). pending = condition not complete at this snapshot, not zero. The current paper tables are a separate, protocol-mismatched reference; do not combine them with this table.

Baseline: 9 previous methods from the strict 5,955 sources; BQ/TC normalized by the actual 5,955 Qcloud/full_cloud_cost.
Repeat identity: SelfAsk/Entropy/MaxLogits/Conf. Margin = 3 fixed executions; Emb. MLP/HybridLLM/RouteLLM/MeanProbe = 3 training seeds; Jev = 3 actual API executions. A reported 0.00 is a measured sample SD for identical fixed executions, not a missing value.
StartLux: one exploratory execution per condition; no SD. pending = condition not complete, not zero.
The paper reference is separate and protocol-mismatched (PAPER_REFERENCE.md); do not combine the two tables or rank them together.

Table 2. BQ, % (higher is better)

	Budget-limited quality: BQ ↑					
Method	**Alpaca**	**MMLU**	**Big-Math**	**Magpie**	**MMLU-Pro**	**MATH**
Strict 5,955 baseline — mean ± sample SD; repeat identity in notes						
SelfAsk	92.21 ± 0.00	80.44 ± 0.00	81.11 ± 0.00	83.43 ± 0.00	67.25 ± 0.00	64.30 ± 0.00
Entropy	91.65 ± 0.00	79.37 ± 0.00	79.20 ± 0.00	81.03 ± 0.00	64.96 ± 0.00	65.54 ± 0.00
MaxLogits	92.56 ± 0.00	78.87 ± 0.00	78.74 ± 0.00	83.10 ± 0.00	64.29 ± 0.00	65.48 ± 0.00
Conf. Margin	91.61 ± 0.00	78.57 ± 0.00	80.06 ± 0.00	81.01 ± 0.00	65.18 ± 0.00	64.49 ± 0.00
Emb. MLP	91.27 ± 0.20	79.79 ± 0.16	78.92 ± 0.35	78.03 ± 0.14	65.37 ± 0.17	65.56 ± 0.13
HybridLLM	91.01 ± 0.07	80.19 ± 0.10	79.68 ± 0.46	78.09 ± 0.89	64.65 ± 1.11	64.95 ± 0.74
RouteLLM	90.96 ± 0.01	81.22 ± 0.05	79.73 ± 0.05	77.80 ± 0.07	69.30 ± 0.24	64.02 ± 0.12
Jev	92.59 ± 0.09	81.66 ± 0.05	79.62 ± 0.07	81.92 ± 0.06	61.62 ± 0.14	62.55 ± 0.08
MeanProbe	91.27 ± 0.03	80.75 ± 0.08	79.90 ± 0.04	78.01 ± 0.07	64.85 ± 0.14	65.81 ± 0.10
StartLux — one exploratory run, no SD						
StartLux 0.8B Instant (Q)	92.34	81.06	79.87	78.87	63.39	64.78
StartLux 0.8B Instant (Q+A)	94.26	82.08	84.91	85.19	66.24	66.68
StartLux 2B Instant (Q)	92.46	82.61	79.62	79.61	65.61	66.15
StartLux 2B Instant (Q+A)	94.03	83.68	84.72	85.80	67.23	66.48
StartLux 4B Instant (Q)	92.30	80.92	79.65	81.84	61.61	63.05
StartLux 4B Instant (Q+A)	95.01	84.75	85.96	86.79	67.32	66.32
StartLux 9B Instant (Q)	91.96	82.03	78.92	82.33	61.93	63.59
StartLux 9B Instant (Q+A)	94.53	85.30	86.28	86.90	67.28	66.25
StartLux 27B Instant (Q)	92.68	80.79	79.33	83.49	62.41	62.49
StartLux 27B Instant (Q+A)	95.21	87.38	86.98	86.97	68.33	65.58
StartLux 9B AutoThink (Q)	92.12	80.64	78.87	82.10	63.96	62.94
StartLux 27B AutoThink (Q)	91.88	81.45	79.79	82.28	61.65	62.54

Strict clean 5,955 comparison: BQ at the middle budget range, as a percentage of all-cloud quality (higher is better). BQ is normalized only by the actual 5,955 clean Qcloud. Baseline: 9 previous methods, SD across the actual repeats named in the table notes. StartLux: one exploratory execution per condition (no SD). pending = condition not complete at this snapshot, not zero. The current paper tables are a separate, protocol-mismatched reference; do not combine them with this table.

Baseline: 9 previous methods from the strict 5,955 sources; BQ/TC normalized by the actual 5,955 Qcloud/full_cloud_cost.
Repeat identity: SelfAsk/Entropy/MaxLogits/Conf. Margin = 3 fixed executions; Emb. MLP/HybridLLM/RouteLLM/MeanProbe = 3 training seeds; Jev = 3 actual API executions. A reported 0.00 is a measured sample SD for identical fixed executions, not a missing value.
StartLux: one exploratory execution per condition; no SD. pending = condition not complete, not zero.
The paper reference is separate and protocol-mismatched (PAPER_REFERENCE.md); do not combine the two tables or rank them together.

Table 3. TC, % (lower is better)

	Quality-first expenditure: TC ↓					
Method	**Alpaca**	**MMLU**	**Big-Math**	**Magpie**	**MMLU-Pro**	**MATH**
Strict 5,955 baseline — mean ± sample SD; repeat identity in notes						
SelfAsk	80.61 ± 0.00	76.25 ± 0.00	73.93 ± 0.00	75.54 ± 0.00	82.81 ± 0.00	80.30 ± 0.00
Entropy	76.83 ± 0.00	78.33 ± 0.00	79.47 ± 0.00	68.09 ± 0.00	80.59 ± 0.00	73.03 ± 0.00
MaxLogits	74.64 ± 0.00	78.46 ± 0.00	79.02 ± 0.00	67.88 ± 0.00	83.76 ± 0.00	73.69 ± 0.00
Conf. Margin	76.51 ± 0.00	78.90 ± 0.00	76.86 ± 0.00	68.37 ± 0.00	80.40 ± 0.00	72.93 ± 0.00
Emb. MLP	81.65 ± 0.24	77.73 ± 1.17	67.09 ± 1.29	87.88 ± 0.39	85.44 ± 1.14	76.51 ± 0.33
HybridLLM	85.07 ± 2.29	76.69 ± 1.13	66.60 ± 1.37	83.04 ± 9.94	87.33 ± 2.02	75.99 ± 3.31
RouteLLM	88.43 ± 0.28	67.11 ± 0.30	68.96 ± 0.24	91.28 ± 0.16	70.21 ± 0.62	76.96 ± 0.18
Jev	68.30 ± 0.58	68.20 ± 1.26	67.42 ± 0.44	64.84 ± 0.10	81.14 ± 0.28	74.79 ± 0.52
MeanProbe	81.12 ± 0.25	72.57 ± 0.15	68.34 ± 0.23	84.91 ± 0.35	85.67 ± 0.02	73.41 ± 0.16
StartLux — one exploratory run, no SD						
StartLux 0.8B Instant (Q)	74.14	75.59	72.25	74.02	81.49	73.03
StartLux 0.8B Instant (Q+A)	52.29	63.80	56.98	53.35	75.93	64.37
StartLux 2B Instant (Q)	78.10	73.70	70.55	75.04	77.61	73.29
StartLux 2B Instant (Q+A)	56.38	59.21	46.23	52.44	67.68	62.34
StartLux 4B Instant (Q)	74.33	66.63	70.69	68.20	81.59	73.13
StartLux 4B Instant (Q+A)	50.05	40.02	37.13	48.07	56.90	57.35
StartLux 9B Instant (Q)	77.64	66.12	67.88	69.46	82.04	76.67
StartLux 9B Instant (Q+A)	56.68	39.23	34.09	51.06	58.47	57.46
StartLux 27B Instant (Q)	69.82	62.67	69.04	62.83	82.49	73.11
StartLux 27B Instant (Q+A)	47.15	31.52	31.47	46.49	47.58	54.35
StartLux 9B AutoThink (Q)	73.06	69.25	70.54	67.73	79.39	75.35
StartLux 27B AutoThink (Q)	78.87	60.51	71.51	72.89	81.46	73.11

Strict clean 5,955 comparison: TC at the middle quality-target range, as a percentage of all-cloud generation cost (lower is better). TC is normalized only by the actual 5,955 clean full_cloud_cost. Baseline: 9 previous methods, SD across the actual repeats named in the table notes. StartLux: one exploratory execution per condition (no SD). pending = condition not complete at this snapshot, not zero. The current paper tables are a separate, protocol-mismatched reference; do not combine them with this table.

Baseline: 9 previous methods from the strict 5,955 sources; BQ/TC normalized by the actual 5,955 Qcloud/full_cloud_cost.
Repeat identity: SelfAsk/Entropy/MaxLogits/Conf. Margin = 3 fixed executions; Emb. MLP/HybridLLM/RouteLLM/MeanProbe = 3 training seeds; Jev = 3 actual API executions. A reported 0.00 is a measured sample SD for identical fixed executions, not a missing value.
StartLux: one exploratory execution per condition; no SD. pending = condition not complete, not zero.
The paper reference is separate and protocol-mismatched (PAPER_REFERENCE.md); do not combine the two tables or rank them together.